TOR AMPLITUDE DESCENDS ALONG THE ETALE LOCALIZATION MORPHISM OF SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof proves one direction by the pullback lemma and the other by tensoring with an arbitrary module, commuting epsilon^* past the derived tensor product via flatness, and applying conservativity to pull back the vanishing

Editorial note. The label lemma-descend-tor-amplitude names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- tor amplitude of pulled back complexes (Cohomology on Sites, Lemma sites-cohomology-lemma-tor-amplitude-pullback)
- flatness of the site morphism epsilon (lemma-epsilon-flat of the same chapter) and the resulting compatibility epsilon^*(E tensor^L F) = epsilon^*E tensor^L epsilon^*F
- the conservative property of epsilon^*: epsilon^*F = 0 implies F = 0 (lemma-epsilon-flat of the same chapter)

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3201; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at proof line(s) 3209; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08H5: sources/stacks-chow/tags.txt, source lines 10768–10768. Prerequisite pointer invoked at proof line(s) 3206; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let E be an object of $D(\mathcal{O}_X)$. Then*

(1) *E has tor amplitude in $[a, b]$ if and only if $\epsilon^* E$ has tor amplitude in $[a, b]$.*
(2) *E has finite tor dimension if and only if $\epsilon^* E$ has finite tor dimension.*

Here ϵ is as in (08H7).

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Proof. The easy implication follows from Cohomology on Sites, Lemma 08H5. For the converse, assume that ϵ^*E has tor amplitude in $[a, b]$. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. As ϵ is a flat morphism of ringed sites (Lemma 08H8) we have

$$\epsilon^*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}) = \epsilon^*E \otimes^{\mathbf{L}}_{\mathcal{O}_{étale}} \epsilon^*\mathcal{F}$$

Thus the (assumed) vanishing of cohomology sheaves on the right hand side implies the desired vanishing of the cohomology sheaves of $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}$ via Lemma 08H8. $\square$